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Carbon credit mechanisms for sustainable agriculture and opportunities in North East India

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Abstract

Greenhouse gas (GHG) emissions are the major cause of global warming, which threatens ecosystems, agricultural sustainability, and human well-being worldwide and thus requires immediate, science-based mitigation measures. Carbon credits have become a vital tool in market-based solutions, increasing the cost of emissions and compensating for emission reductions, thereby promoting environmentally friendly technologies and creating economic opportunities that support livelihoods based on agriculture. In developing countries such as India, agriculture plays a dual role: both a significant emitter and a potential carbon sink, capable of contributing meaningfully to climate change mitigation and rural development. Using an integrated review-based analytical framework grounded entirely in secondary data, the paper critically examines greenhouse gas emission trends, carbon market architectures, biophysical carbon quantification models, and economic valuation approaches in India, with particular emphasis on the North Eastern Region (NER). The study examines the phases of carbon generation, carbon pricing frameworks, and the integration of sustainable agricultural practices that allow farmers to access carbon markets. The case studies, such as the Khasi Hills Community Carbon Project and the EKA initiative, illustrate how carbon farming initiatives are designed to generate measurable emission reductions under established verification frameworks. However, limitations include monitoring, reporting, and verification (MRV) constraints, policy gaps, and limited farmer awareness. The study highlights that carbon credit mechanisms have the potential to support emission reduction while contributing to agricultural sustainability, with implications for farm-level income diversification, low-carbon farming, and sustainable economic growth in the Indian agricultural sector.

Keywords Agricultural sustainability, Carbon credits, Carbon markets, Carbon sequestration, Greenhouse gas emissions, Sustainable development goals

1 Introduction

Nowadays, our planet is heating [1], and climate change has become an inevitable reality [2]. Climate change is rapidly intensifying, which is projected to exceed the Paris target of a global warming limit of 1.5 °C by 2035 [3]. Sustainable prevention involves



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reducing GHG emissions, especially by lowering atmospheric CO₂ concentrations [4]. To address this, carbon credits have emerged as a vital tool for transitioning to a low-emission pathway to combat global warming [5], which is technically feasible [6] and economically viable [7]. A carbon credit may represent either an emission allowance under compliance markets or an offset generated through verified emission reduction, carbon sequestration activities, depending on the regulatory context [8] and is tradable in carbon markets [9]. The idea of carbon credits was proposed in 1997 and introduced in the Kyoto Protocol, which entered into force in 2005 [10]. Aiming to reduce GHG emissions through carbon trading, the Paris Agreement reinforced the notion of carbon credits in 2015 [11]. Union Minister of Power, Manohar Lal, confirmed that India will start its first carbon credit trading scheme in mid-2026 [12]. Increasing evidence shows that agriculture can be essential in removing GHGs from the atmosphere [13]. Biophysical models have been widely applied for carbon quantification in agriculture [14], offering frameworks to establish baselines [15], assess sequestration potential [16], and track emission reductions across diverse land use systems [17]. These models, coupled with the institutional frameworks of carbon markets that are broadly categorized into compliance carbon markets (CCMs) [18], which operate under legally binding emission reduction obligations, and voluntary carbon markets (VCMs) [19], where emission reductions are traded voluntarily by firms and institutions. In India, agricultural mitigation initiatives, particularly in the North Eastern Region, are presently more aligned with voluntary carbon markets due to methodological flexibility, lower transaction thresholds, and the predominance of smallholder and community-managed land systems. Generation phases and stakeholder participation, from verification agencies to local farmer cooperatives, determine the efficiency of credit creation and exchange. Importantly, carbon finance provides farmers with additional revenue streams, aligning ecological benefits with socio-economic incentives. Hence, by granting incentives to farmers for their low-carbon farming practices, we can trap carbon emissions, which lead to revenue generation for the farming community. By providing financial incentives for verified emission reductions, carbon credit mechanisms can align climate mitigation objectives with farmers' economic interests; this study analytically examines this alignment in the Indian context, with particular emphasis on the North Eastern Region, using evidence from existing studies, market frameworks, and documented case projects.

2 Conceptual and analytical framework

2.1 Greenhouse gas (GHG) emissions: global and Indian agricultural context

Greenhouse gases include chlorofluorocarbons (CFCs), ozone, nitrous oxide, methane, and carbon dioxide, which significantly absorb infrared radiation [20]. The greenhouse gas emission trends from 1990 to 2025 are illustrated in (Fig. 1). These emissions enhance the greenhouse effect, leading to global warming [21], adverse health effects [22], and environmental degradation [23].

2.2 GHG emission trends

The greenhouse gas (GHG) emissions from land use and fossil fuels have risen gradually, peaking in 2019 [25]. The intergovernmental panel on climate change (IPCC) working group III (WG3) recognized five major economic sectors as sources of global GHG emissions: energy systems, industry, buildings, transportation, and AFOLU (agricultural,

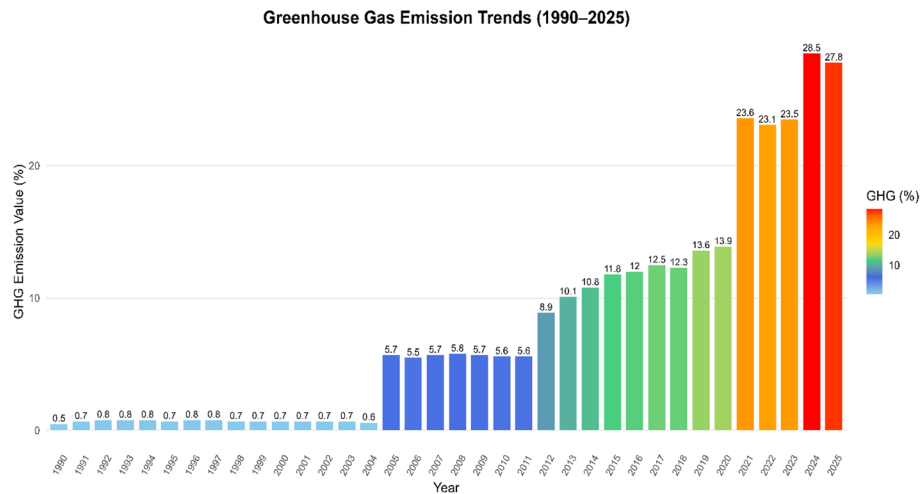


Fig. 1 GHG emission trends (1990–2025) (Data Source: [24])

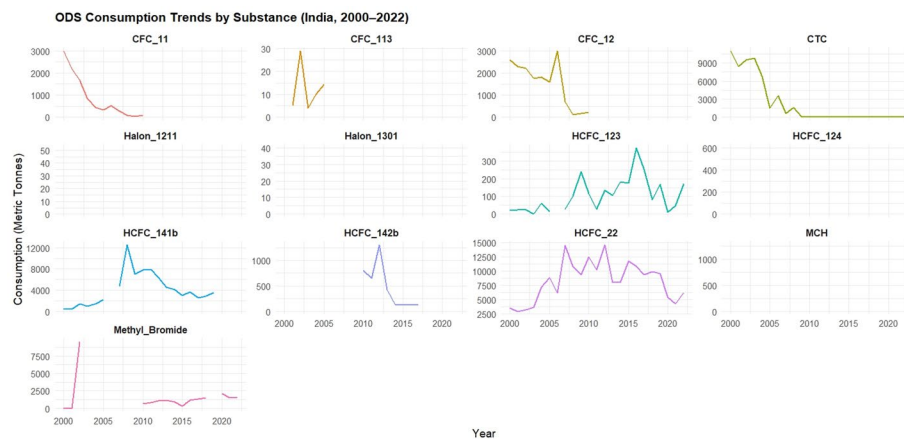


Fig. 2 ODS consumption trends by various substances in India (2000–2022) (Data Source: [31])

forestry, and other land uses) [26]. The GHG emissions level in 2018 reached a historical peak of approximately 5.8 GtCO₂eq, which is 11% higher than in 2010 [27]. The energy systems sector accounted for the greatest portion of emissions in 2018 (34%; 20 GtCO₂eq), followed by industry (24%; 14 GtCO₂eq), AFOLU (21%; 12 GtCO₂eq), transportation (14%; 8.3 GtCO₂eq), and building operations (6%; 3.3 GtCO₂eq) [28]. India's GHG emissions increased by 6.1 per cent in 2023 compared to 2022. In 2023, the shares of CO₂, CH₄, F-gases, and N₂O in total national emissions expressed in CO₂eq were 71.5%, 20.3%, 1.7%, and 6.5%, respectively [29]. Seventy per cent of India's CO₂ emissions are attributed to these six sectors: power, steel, automotive, aviation, cement, and agriculture [30]. The consumption trends of Ozone Depleting Substances (ODS) by type in India from 2000 to 2022 are shown in (Fig. 2).

2.3 Agricultural contribution to GHGs

According to Gorain et al. [32], total emissions in 2018 (CO₂eq) from the agricultural sector in India are estimated to be 647,814.77 gigagrams. In 2019, the agriculture sector shared 22% of global greenhouse gas emissions [33]. Panchasara et al. [34] report that deforestation and agriculture account for around 30 percentage of greenhouse

gas emissions. A significant amount of GHG emissions are from field crops [35], like paddy [36], wheat [37], and maize [38], and from high-emission farming practices [39]. Biomass burning, rice production, and improper livestock management contribute to 12%, 11%, and 7% of GHG emissions, respectively [40]. The emission intensity for kg^{-1} of bovine meat is 108.3 $\text{kg CO}_2\text{-eq}$ in India [39]. Livestock account for over 70% of all greenhouse gas emissions from agriculture [34]. In 2021, among the primary agriculture sectors, livestock farming contributed more than half of the GHG emissions, followed by crop production, and on-farm fuel emission share of 28% and 21% respectively, of the agricultural emissions [41]. Bonesmo et al. [42] say about 6.1 gigatonnes of carbon dioxide equivalents ($\text{CO}_2\text{-eq}$) per year are emitted from agriculture, representing 10 to 12 per cent of global GHG emissions. Cariappa (2023) [43] reveals that GHG emissions from agricultural residue burning in India have increased by 75% since 2011. In 2023, the majority of GHG emissions consisted of CO_2 [29]. With a share of approximately 7.8 per cent of the total global emissions in 2023, India is the third-largest emitting economy after China and the United States [44]. By 2030, it is anticipated that the agriculture sector will generate between 35 and 60 per cent more nitrous oxide (N_2O), which can be attributed to higher nitrogen content from fertilizer use and increased production of waste from livestock [45]. Hence, the cost-effective reduction from this sector could significantly reduce India's total emissions. Therefore, carbon credits represent a sustainable solution to combat climate change by minimizing GHG emissions [46].

2.4 Scope for decarbonization in India

The 26th conference of the parties (COP26) paved the way for India's decarbonization path [47] to achieve a 50 per cent reduction in its energy emissions by 2030 [48]. The Prime Minister of India declared at COP26 that India would reach net-zero emissions by 2070 [49]. Achieving global Net Zero is fundamental to ensuring economic stability, safeguarding human health, and protecting the environment [50, 51]. The pursuit of rapid economic growth often stands in contrast to the objectives of rapid decarbonization [52], making it challenging to meet these twin objectives [53]. As a developing country, India ratified the Kyoto Protocol on 26th August 2002 and is responsible for its share in the mitigation of global GHG emissions [54]. Previous analyses suggest that carbon trading mechanisms may create additional foreign exchange opportunities, contingent upon market participation and pricing stability [32]. India's net-zero path can be achieved by current policies, accelerated technology adoption, carbon pricing, carbon capture, usage, and storage (CCUS) technologies [55], which enables the potential of India to create 287 gigatons of carbon space for the world, which accounts for half of the global carbon budget [56]. Mitigation of greenhouse gas (GHG) emissions within the land use, land-use change, and forestry (LULUCF) sector in India can be effectively pursued through the enhancement of soil organic carbon in agricultural lands [57]. Projections indicate that GHG emissions from the agriculture, forestry, and other land use (AFOLU) sector are expected to decline more sharply than those from other sectors, with an estimated reduction of 39% between 2020 and 2050 [58]. Notably, cropland and afforested areas serve as key sites for carbon sequestration, while grasslands function as increasingly emphasized carbon sinks [59]. Lal and Logan [60] stated that about 15 per cent of global GHG emissions can be offset by agricultural soils. Motta et al. [61] report that India has recently become a world leader in reducing greenhouse gases by

adopting the CDM. Although region-specific emission inventories for the North Eastern Region remain limited beyond 2020, recent projections for 2024–2025 suggest continued growth in agricultural greenhouse gas emissions, particularly N_2O from fertilizer use and methane from irrigated paddy systems. Scenario-based projections using DNDC and FAO growth trajectories indicate a potential 8–12% increase in agricultural emissions by 2025 in the absence of mitigation interventions [13]. Cumulative carbon certified emissions reductions (CERs) issued to CDM projects in India are depicted below (Fig. 3). The stabilization in recent years indicates a maturity of existing projects but also highlights the need for diversification into new avenues of decarbonization.

2.5 Significance of carbon credits

A carbon credit is a certificate showing that a government [63] or company [64] has paid to have a certain amount of carbon dioxide removed from the environment [65]. According to Kaisto [66], carbon credits are permits that allow the owner to emit a certain amount of carbon dioxide or other greenhouse gases (GHGs). The carbon credit system is a recent market-based emission reduction mechanism designed to mitigate climate change by quantifying carbon reductions and rewarding individuals [67, 68]. Carbon credits are part of market-based solutions to reduce greenhouse gas emissions, permitting the holder to emit a certain amount of carbon dioxide or other greenhouse gases [69], to incentivize emission reductions through economic mechanisms [70]. The cap-and-trade model that was used to minimize sulphur pollution in the 1990s is the basis for carbon credits [71]. India issued 35.91 million carbon credits from 2010 to 2022 and traded them on global markets. India has emphasized the creation of a domestic carbon market to meet its COP-26 commitments and to fulfil its Nationally Determined Contributions [72].

According to the Environment Protection Authority Victoria [73], carbon credit refers to a standardized unit representing the reduction of greenhouse gas (GHG) emissions, conventionally quantified as the equivalent of one metric ton of carbon dioxide (CO_2) mitigated, i.e., 1 carbon credit = 1 ton of carbon dioxide reduction. Carbon offset equivalence is a concept to measure and compare the reduction of greenhouse gases (GHG) [74] from different mitigation actions [75], and expressed in carbon dioxide equivalent (CO_2e) [76]. Emission reductions achieved through renewable energy deployment,

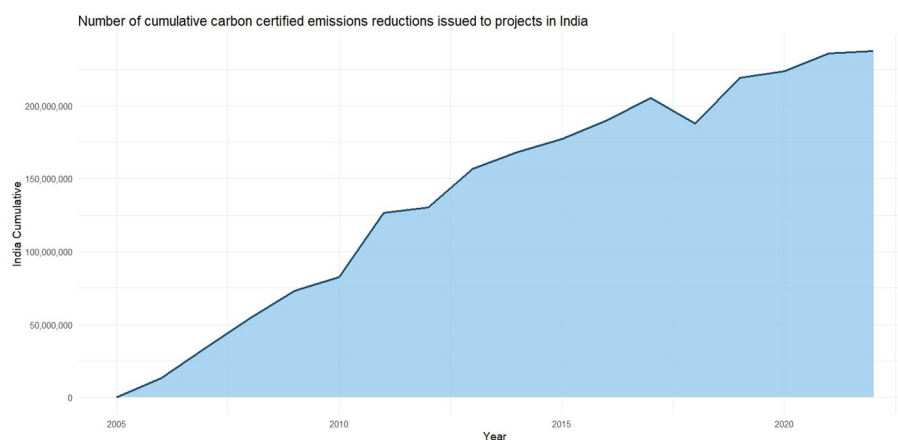


Fig. 3 Number of cumulative carbon CERs issued to projects in India (Data Source: [62])

afforestation initiatives, and climate-resilient agricultural management practices are quantified into standardized carbon equivalents, which are subsequently traded within carbon markets [77, 78]. Monitoring, reporting, and verification (MRV) will be used to measure these carbon offsets, ensuring additionality [79] and permanence [80] while preventing leakage, which is the displacement of GHG emissions to other locations [81]. IPCC-approved protocols [82] and life cycle assessment (LCA) [83] are used to evaluate these measurements across the sectors and projects. The extent of equivalence is, however, challenged by temporal discrepancies such as direct fossil emissions [84] and uncertainties about permanence [85]. According to Cabiyo and Field [86], carbon credit pathways can be constrained by optimistic assumptions, leading to over-credit [87] and a loss of environmental integrity [88]. Empowering the MRV systems, implementing conservative baselines, and integrating risk-buffering mechanisms are imperative to ensure the authenticity and permanence of carbon credits.

3 Carbon markets and institutional architecture

3.1 Carbon credit markets

Carbon credits are generated, transacted, and exchanged within a regulated trading system known as the carbon credit market [89], wherein companies and individuals acquire credits from entities that have demonstrably reduced greenhouse gas emissions [69]. This market approach achieves balance by enabling emitters to offset their greenhouse gas emissions through the purchase of credits from actors engaged in verified carbon sequestration or emission reduction activities [90]. The carbon credit market will promote climate action by providing incentives for transferring carbon credits and financial resources from one entity to another to support activities that effectively mitigate greenhouse gas (GHG) emissions [91]. The value of the voluntary carbon market (VCM) activity increased more than tenfold between 2017 and 2021 [92]. Due to the development of several carbon pricing instruments and biophysical models, as well as the carbon offsetting and reduction scheme for international aviation (CORSIA), there will be increased demand from compliance carbon markets (CCMs) shortly [93]. Table 1 presents representative biophysical models commonly used for quantifying agricultural carbon. These models have been applied in heterogeneous agroecosystems and may be adaptable to smallholder contexts, subject to calibration and local data availability. The table is intended to illustrate methodological options rather than provide an exhaustive comparison.

3.1.1 Digital measurement, reporting and verification (digital MRV)

Recent advances in digital monitoring technologies have significantly transformed agricultural carbon accounting. Remote sensing (RS), geospatial analytics, and machine learning approaches are increasingly integrated with biophysical models such as DNDC and CENTURY to enhance measurement accuracy and reduce verification costs. Satellite-based vegetation indices and radar imagery enable estimation of biomass dynamics and land-use change across heterogeneous landscapes [17]. Digital MRV systems improve scalability by reducing reliance on costly field-based soil sampling [101], particularly in fragmented and topographically complex regions such as the North Eastern Region (NER) of India. Studies demonstrate that combining process-based models with earth observation data enhances spatial resolution and uncertainty reduction in soil

Table 1 Biophysical models for carbon quantifications

Model name	Purpose	Key features	Application in carbon farming	How it helps in carbon credit	References
CENTURY	Simulates long-term soil organic matter and nutrient dynamics	Models C, N, P, and S cycling in crops and grasslands	Estimates the SOC sequestration over decades under different land uses	Calculates baseline and project soil carbon stock changes for carbon credit quantification	[94]
DNDC	Simulates biogeochemistry in an agroecosystem	Tracks GHGs (CO ₂ , N ₂ O, CH ₄), crop growth, soil C dynamics	Evaluates GHG mitigation practices in agriculture	Quantifies GHG reductions eligible for carbon crediting under protocols like CDM or VCS	[95]
RothC	Models SOC turnover in non-flooded soils	Simple input needs; long-term soil C predictions	Estimates SOC accumulation with organic inputs and conservation tillage	Converts SOC changes into CO ₂ e credits for land-based offset projects	[96]
DAYCENT	The daily version of CENTURY	Daily weather, GHG fluxes, crop-soil interactions	High-resolution GHG flux modelling	Provides daily GHG reduction data to enhance precision in carbon credit MRV systems	[97, 98]
APSIM	Models biophysical farm systems	Modular; simulates crops, climate, soil	Evaluates trade-offs in carbon and yield under different practices	Verifies soil and biomass carbon changes for credit issuance and scenario analysis	[99]
EPIC	Assesses crop yield and environmental impacts	Includes erosion, nutrients, water, yield, and carbon	Policy and scenario evaluation for sustainable farming	Projects carbon offset potential from land-use changes and sustainable farming practices	[100]

carbon estimation [14, 16]. For smallholder-dominated agro-ecologies, digital MRV can substantially lower transaction costs and improve transparency, thereby strengthening environmental integrity in voluntary and compliance carbon markets.

3.2 Market structure: CCMs and VCMs

To reduce GHG emissions, carbon markets set a price for carbon emissions and reward stakeholders who sell carbon credits [102]. Many national governments and private corporations rely on carbon markets to meet emission reduction targets and pursue carbon neutrality [103]. Regime interactions underlying the causal mechanism can be classified into four types, namely cognitive, rule-based, behavioural, and functional interactions [104]. These interactions predominantly occur in the direction of voluntary carbon markets (VCMs) influencing compliance carbon markets (CCMs), with VCMs characterized as ‘soft’ regimes and CCMs as ‘hard’ regimes, as observed by Chen and Xie [19]. A deeper understanding of the relationship between CCMs and VCMs is important for policy frameworks (Fig. 4).

3.3 Carbon finance for sustainable agriculture

Under Article 6 of the Paris Agreement, an international regulatory framework is established among participating countries [108], engaging them in carbon trading to reach mitigation targets by 2030 to achieve Nationally Determined Contributions (NDCs) [109]. Under Article 6.2 of the Paris Agreement, countries can trade “Internationally Transferred Mitigation Outcomes” (ITMO) to balance the climate targets [110]. Gold Standard supports the land-use project and Voluntary standards like Verra (VCS) in compliance with the Article 6 label, introducing projects like Rwanda’s cookstoves [111]. Positioning with the Paris Agreement enables global financial institutions to facilitate

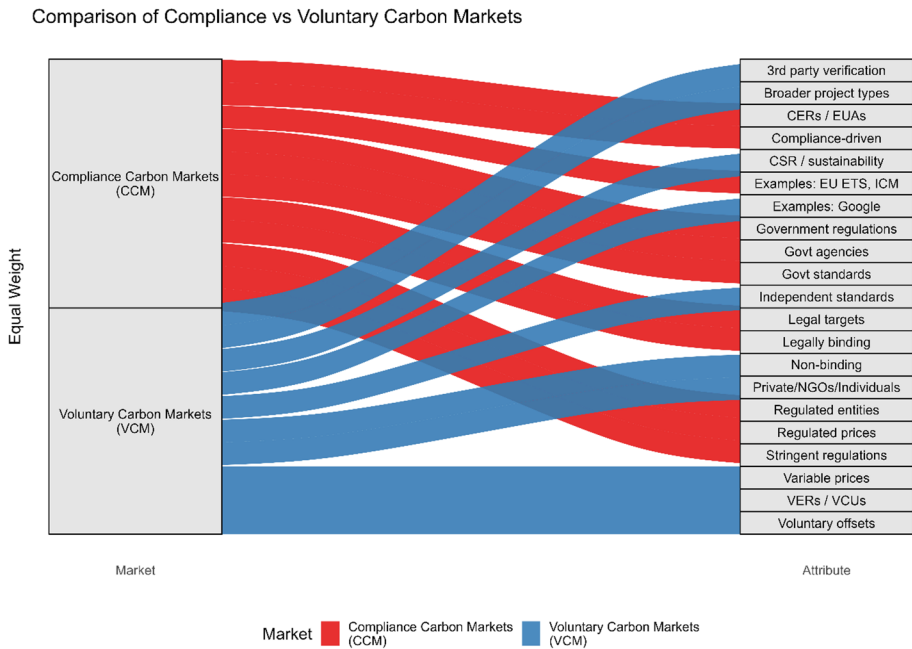


Fig. 4 Comparison between CCM and VCM [105–107]

the transition toward a low-carbon agricultural economy [112] and alleviate credit risks linked to climate-related events [113]. According to the Network for Certification and Conservation of Forests [114], India’s Carbon Registry (CRI), launched in 2021, issues Indian carbon units to certify, record, and facilitate the trade of verified GHG emission reductions. Linking carbon markets to farm production and incentivizing emission reduction with the help of tradable carbon credits is lauded as a win–win strategy [115]. There is a significant potential for developing ITMO-aligned projects in the North East Region (NER) of India through afforestation, agroforestry, and soil carbon enhancement initiatives [116]; this development aligns with carbon market frameworks. These projects enable monetization of carbon sequestration, which presents co-benefits such as conservation of biodiversity and diversification of rural livelihood. Although NER is ecologically vulnerable and socio-economically susceptible, ITMOs present a strategically feasible means of directing climate finance and facilitating sustainable development [117]. This economic mechanism paves the way for a promising approach to earn carbon credits and neutralizing carbon emissions.

3.4 Indian carbon market framework

The central government is the authority that issues carbon credit certificates (CCCs) [118]. The Carbon Credit Trading System (CCTS) framework is a system on which these CCCs are exchanged [119], a system administered by the Bureau of Energy Efficiency (BEE), under the Ministry of Power, and establishes rules and regulations on carbon trading [120]. This carbon market framework facilitates the transition to a low-carbon economy. This system is monitored by the National Steering Committee on Indian Carbon Market (NSCICM), along with the Bureau of Energy Efficiency (BEE), to guarantee its integrity, credibility, and trustworthiness [121].

3.4.1 Institutional setup (BEE, NSCICM)

In India, the Bureau of Energy Efficiency (BEE) contributes to agricultural decarbonization through initiatives promoting energy-efficient irrigation pumps, biomass-based energy use, and improved on-farm energy efficiency [122]. These interventions indirectly support emission reduction goals and enhance the eligibility of agricultural activities for participation in domestic and voluntary carbon markets. Li and Colombier [123] pointed out that energy efficiency in construction and its upgrade is promoted by using both regulatory measures and cash incentives. BEE oversees and supports the sector's energy efficiency policies, ensuring that required standards and codes are met and continually improved [124]. By regulating, rewarding, and monitoring, BEE aims to improve energy efficiency and support the creation of carbon credits. The combination of these measures will help India move closer to its climate goals and create energy-saving projects that will be funded through carbon markets.

Simplified rules, effective institutions, and enforced environmental policies are needed to stabilize carbon markets in India to achieve sustainable results [125]. The National Steering Committee for Indian Carbon Market (NSCICM) will supervise the progress and operation of carbon credit projects and a marketplace for trading carbon certificates in India. It ensures that carbon credit trading in India aligns with the country's climate change goals [126]. The Central Government has set up the NSCICM as part of the Carbon Credit Trading Scheme (CCTS), by which NSCICM will operate the Indian Carbon Market (ICM). This committee has members across Ministries and related organizations, with the Secretary of the Ministry of Power serving as Chairperson and the Secretary of the Ministry of Environment, Forest and Climate Change serving as Co-Chair [127].

3.4.2 CCCs and trading framework (CCTS)

Global efforts to reduce greenhouse gas emissions depend on carbon credit certifications and trading systems. Addressing emissions and enhancing a sustainable economy, stakeholders can now deal with carbon credits [128]. The Indian Carbon Market Framework has two key mechanisms: the Compliance Mechanism and the Offset Mechanism [127]. The design, certification, and operation of carbon credits and trading systems are becoming increasingly effective. The application of different policies, standardized certifications, and advanced digital tools enhances transparency and supports global emission reduction efforts [129].

3.4.3 Evolutionary timeline of the Indian carbon market

In 2023, India shifted its focus on legislation to the institutional operationalization after the announcement of the Carbon Credit Trading Scheme (CCTS). The Bureau of Energy Efficiency [130] completed the MRV framework, accreditation process, and guidelines on the offset mechanism in 2024–2025, and the sector target, along with the final starting point of emission intensity, was also announced under the GEI Rules 2025. These measures provided a legal, technical, and market infrastructure needed for compliance trading. The Indian Carbon Market became market-ready by 2026, with targets formalized and exchange-based trading confirmed (Table 2).

Table 2 Chronological evolution of India’s carbon credit trading scheme (CCTS) and associated regulatory framework (2022–2026)

Year	Key developments (Policy/Institutional)	References
2022	The Energy Conservation (Amendment) Act, 2022 (Act No. 19 of 2022) was enacted, empowering the central government to specify a carbon credit trading scheme under the Energy Conservation Act	[131]
2023	Ministry of Power notification S.O. 2825(E) establishes the Carbon Credit Trading Scheme, 2023 (CCTS), creating a compliance mechanism with GHG-intensity targets. CCTS amended (S.O. 5369(E)) to insert an “offset mechanism” for voluntary (non-obligated) entities	[132]
2024	The Bureau of Energy Efficiency (BEE) publishes the detailed procedure for the compliance mechanism (MRV framework) and related accreditation rules. These finalized guidelines (issued in July 2024) operationalize GHG-intensity measurement/reporting and verifier accreditation	[133]
2025	BEE issues the Detailed Procedure for the Offset Mechanism (Version 1.0), effective from March 27, 2025. Ministry of Environment, Forest and Climate Change notifies the Greenhouse Gases Emission Intensity Target Rules, 2025 (G.S.R. 739(E)), prescribing mandatory sectoral emission-intensity targets for obligated entities	[134, 135]
2026	Government confirms that GEI targets have been notified for the identified energy-intensive sectors and that carbon credit certificates will be tradable on power exchanges, completing the regulatory framework for market launch	[136]

4 Carbon credit generation pathways in agriculture

4.1 Generation phases and stakeholders

The carbon credit markets have occurred in distinct phases [91], and each phase brings new challenges and improvements through regulatory frameworks, market mechanisms, and policy goals [137], to ensure market credibility, efficiency, and effectiveness. Through these phases, compliance and voluntary carbon markets are maturing, together with national and multilateral strategies targeted at decreasing carbon emissions [138]. Understanding the historical evolution of carbon markets alongside diverse roles and interactions of stakeholder roles is essential to comprehending their operational dynamics and prospective trajectories of development [91]. The following figure (Fig. 5) schematizes stakeholders’ key roles in carbon credit markets.

4.2 Project and market development phases

This phase involves project initiation, feasibility analysis, and market guidance. It also includes strategic project shaping, stakeholder engagement [142], and analysis of emission reductions compared with traditional practices [143]. Project developers assess the land-use potential along with supplemental attributes such as additionality beyond baseline practices, permanence, and co-benefits, such as biodiversity conservation and rural income enhancement [144]. Financial modelling is undertaken to evaluate project viability and cost-effectiveness concerning prevailing carbon market prices [145]. Their regulatory consistency is with national and international carbon standards (Verra, Gold Standard), upon which they are certified and traded [146]. Risk assessments addressing non-permanence, leakage, and policy uncertainty are also incorporated into project design. The use of digital MRV tools and blockchain-based registries ensures streamlined, transparent, and traceable credit issuance and transactions. Carbon credit buyer guidance follows four phases: defining quality, developing and testing assessment methodologies, applying it to various credits, and finally integrating results and recommendations [147]. These integrated efforts improve project credibility and also foster scalable, replicable carbon finance models.

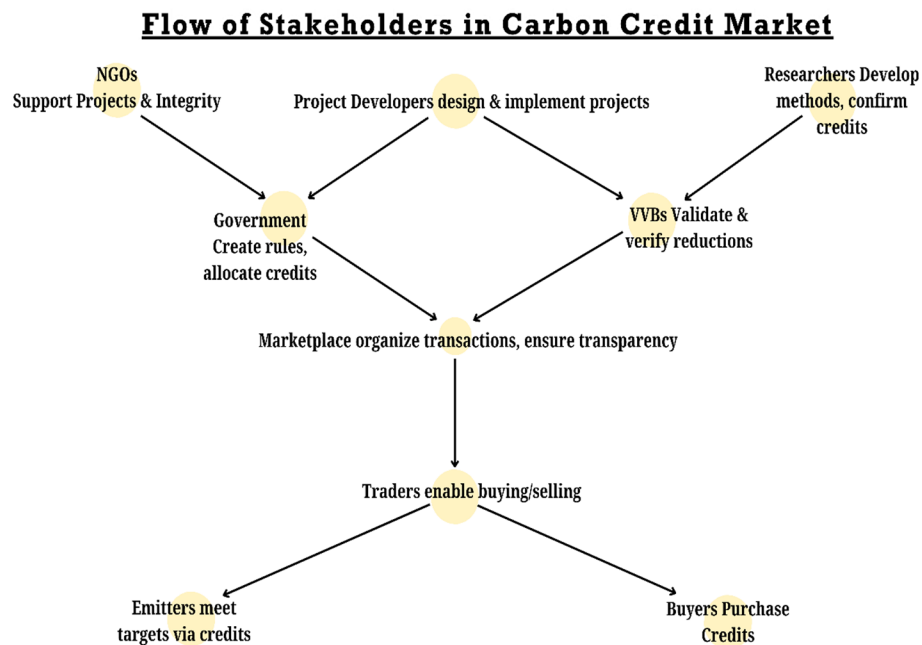


Fig. 5 Stakeholders and their roles in carbon credit markets [139–141]

4.3 Assessment and verification phases

Carbon credit assessment deals with baseline emissions and determines eligible activities resulting in emission reductions [147]. Subsequently, the project is evaluated using internationally recognized methodologies such as the clean development mechanism (CDM) and verified carbon standard (VCS) [148]. A comprehensive project design document (PDD) is prepared, outlining the project scope, technologies used, and estimated carbon savings [149]. This is followed by the verification step, during which emission reductions are confirmed by a designated operational entity (DOE), an independent auditor accredited by the CDM Executive Board, who cross-checks actual data with the baseline estimate [150]. This process includes on-site visits, data audits, and consultations with interested parties to ensure credibility and transparency. Once the verification is successful, carbon credits are issued and certified.

4.4 Policy and regulatory phases

The policy and regulatory cycles in carbon credit systems follow structured stages to ensure environmental integrity and efficiency in the market [88]. The policy formulation phase determines the targets to be achieved through emission reductions under international agreements such as the Paris Agreement, priority sectors, and establishes baselines [91]. Following this, the regulatory development phase creates legal provisions outlining the scope and eligibility of credits and determining whether credits fall under compliance or voluntary markets [151]. Finally, an enforcement and review phase determines credit issuance based on verified reductions. This entails institutionalization of MRV (measurement, reporting, and verification), accreditation procedures, and transparency standards [152]. In developing countries, co-benefits such as rural livelihoods and ecosystem services are also emphasized in policies. Effective regulation ensures stakeholder confidence, participation, and overall climate impact.

5 Socio-economic and environmental outcomes

Carbon credits, in addition to offsetting greenhouse gas (GHG) emissions, also generate substantial socio-economic benefits and enhance ecosystem integrity [153]. According to Ghosh et al. [154], the promotion of sustainable farming practices enhances the generation of carbon credits while simultaneously supporting rural livelihoods. In the predominantly agrarian Northeast, carbon-credit projects connect climate action with local development [155]. These initiatives align with SDG 13 (Climate Action) by rewarding farmers and communities for sequestered CO₂, thereby improving livelihoods and contributing to SDG 1 (No Poverty) and SDG 2 (Zero Hunger). Agroforestry systems improve soil carbon and fertility, thereby enhancing food security and farm income [156]. Therefore, carbon finance provides alternative sources of income to farmers, thereby reducing poverty and enabling ecosystem regeneration. In India's North Eastern Region (NER), implementing carbon credit mechanisms provides farmers with opportunities to diversify income streams, thereby enhancing rural livelihoods, while aligning with national objectives for long-term environmental sustainability. Table 3 represents the carbon credits aspects aligning with the UN SDGs.

5.1 Gender, livelihood, and community development

In Northeast India, jhum cultivation remains a vital source of rural livelihood [166]. Carbon payment schemes aimed at conserving and enhancing forest resources channel financial resources directly to local communities [167], thereby complementing subsistence-based livelihoods with diversified income opportunities [168]. Agroforestry on degraded land (trees over crops) enhances farm productivity and generates marketable timber and fruit, further increasing household income. Singh et al. [169] report that, in the low-land semi-arid region of the Vindhyan ranges, practising agroforestry systems, such as guava-based and mango-based agri-horticultural systems, as well as pure orchards of guava, mango, Indian gooseberry, and teak boundary plantations, can earn an additional 7% of the total revenue through carbon credits. Many carbon projects organize village cooperatives or forest committees to manage funds. Revenues support community welfare by establishing wells, schools, clinics, and group enterprises, thereby building local institutions [170]. By facilitating collective land-use transitions, carbon initiatives contribute to strengthening social capital and enhancing governance capacities within tribal and rural communities [171]. Women's empowerment in the North Eastern Region (NER) of India can be achieved through inclusive project design within the carbon credit context. Traditionally, women are the primary stewards of fuelwood and livestock management; implementing clean-energy solutions such as biogas and efficient cookstoves alleviates their labour demands and health constraints, while rewarding them with carbon credits [161]. The W+ credit structure, which measures and rewards women's empowerment [172], ensures their participation in decision-making and equitable revenue sharing, and project-based training further strengthens their economic agency [173]. Carbon credits thus serve as a mechanism to promote gender equality (SDG 5) and poverty reduction. Tang et al. [174] observed that carbon credit programs provide an additional revenue stream for farmers and open greater opportunities for climate traders and aggregators. Also, White [175] observed that, on the social outlook, carbon credit enhances farmer livelihoods and women's empowerment, fosters community development, and expands market access for green products. Thus, Carbon-credit

Table 3 Carbon credit mechanisms in Northeastern India align with the SDGs

North East- ern States of India	Carbon credit source and mechanism	Income generation potential	Key target groups	Relevance to SDGs	Aligned SDGs	Ref- er- ence
Arunachal Pradesh	The state-operated 3 megawatt (MW) Sumbachu small-scale hydropower project (SHP) has been accredited under a voluntary carbon certification framework	Issued 16,326 tCO ₂ credits by mid-2023, rising toward 97,783 t by 2028, implying significant revenue potential through voluntary market sale	Local Indigenous Communities	Supports climate mitigation by harnessing renewable energy and generating clean energy, providing revenue for the public sector. Enhances the long-term sustainability of power utilities	SDG 13 (Climate Action), SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure)	[157]
Assam	Agroforestry and tree-planting schemes implemented through the NABARD-NABCONS partnership, alongside a TED-led 1 crore tree plantation initiative	Framework signed in 2024 to enable smallholder farmers to earn and sell carbon credits via sustainable agroforestry. For example, Assam's plan to plant 1 crore trees is expected to generate tradable carbon offsets after 3 years. (quantitative value depends on future credit volumes and market prices)	Smallholder farmers and rural communities practising agroforestry, as well as agricultural boards and local NGOs supporting tree cover	Encourages sustainable agriculture, promotes soil health and supports forest cover. Sequestering CO ₂ via trees combats climate change and provides new farmer income, contributing to rural development	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 1 (No Poverty)	[158, 159]
Manipur	Community forest conservation initiatives	Focuses on awareness and capacity-building. Local groups are exploring forestry carbon schemes; however, no quantified revenue has been documented yet	Indigenous Civil Society Organizations and village councils, including representatives from the Rongmei Naga community, engaged in learning about forest carbon credits	Emphasizes forest protection and cultural values. Links forest conservation to carbon finance by valuing ecosystems and addressing GHG emissions	SDG 15 (Life on Land), SDG 13 (Climate Action)	[160]

Table 3 (continued)

North East- ern States of India	Carbon credit source and mechanism	Income generation potential	Key target groups	Relevance to SDGs	Aligned SDGs	Ref- er- ence
Meghalaya	Khasi Hills REDD+ community project, Plan Vivo–certified, protecting 27,000 ha of forest	Through 2022, the project disbursed US\$472,984 to 7,764 local families and invested \$298,817 in forest management. It has avoided more than 411,000 tCO ₂ , demonstrating significant revenues and conservation payments	Khasi tribal village councils and women-led community groups (matrilineal societies) directly benefit from carbon payments	Protecting forest carbon stocks preserves biodiversity while providing income and livelihood support. The project also promotes gender equity by channelling funds through women's institutions	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 1 (No Poverty), SDG 5 (Gender Equality)	[161]
Mizoram	Bamboo-based agroforestry and small-scale bamboo industries	Bamboo sequesters CO ₂ as it grows, providing substantial carbon-storage potential in Mizoram. While specific carbon-market projects are not yet documented, the development of bamboo plantations and handicraft industries generates rural income	Rural households and artisans in bamboo-rich areas, such as villages like Tanhril, where residents produce bamboo products	Bamboo agroforestry supports sustainable rural livelihoods and forest carbon sinks, contributing to poverty reduction and climate action. It also promotes biodiversity-friendly forestry	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 1 (No Poverty), SDG 8 (Decent Work and Economic Growth)	[162]
Nagaland	REDD+ forest-conservation project	Under this 50-year REDD+ initiative, forest communities will trade carbon credits on the voluntary market, generating potential revenue	Indigenous tribal villagers, e.g., in the Pughoboto area, and their local councils, who conserve forests in exchange for carbon revenues	Conserving forests directly reduces CO ₂ emissions while protecting biodiversity. Revenue generation creates sustainable incomes for tribal communities, aiding poverty reduction	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 1 (No Poverty)	[163]

Table 3 (continued)

North East- ern States of India	Carbon credit source and mechanism	Income generation potential	Key target groups	Relevance to SDGs	Aligned SDGs	Ref- er- ence
Sikkim	Natural Climate Solutions (NCS) pathways, including REDD+, agroforestry, and reforestation, are being evaluated by the state	This study identifies several cost-effective options, such as reforestation of fallow land, agroforestry, and protected forests. Potential carbon credit volumes are high, though monetization is under planning	State Forest Department and rural communities, including farmers with fallow lands, are benefiting from planned agroforestry and reforestation projects	Sikkim's high forest cover enhances forest carbon sinks and conserves ecosystems. NCS projects also support rural livelihoods and sustainable land use	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 2 (Zero Hunger), SDG 1 (No Poverty)	[164]
Tripura	Agroforestry on forest land, funded by JICA under the Right of First Refusal (RoFR) patta lands, in conjunction with MGNREGA schemes	The state plans 5,000 ha of plantations; co-benefits explicitly include carbon sequestration. Actual carbon revenue is prospective and contingent on future REDD+ and offset sales	Forest-rights holders and RoFR patta land holders benefit from agroforestry incentives. Government agencies, including the Forest Department and the JICA project, facilitate implementation	Tree-planting schemes increase carbon sinks, restore ecosystems, improve farm income, and foster sustainable agriculture	SDG 13 (Climate Action), SDG 15 (Life on Land), SDG 1 (No Poverty), SDG 2 (Zero Hunger)	[165]

programs strengthen the agrarian economy and the social network of NER by integrating climate finance with livelihood support to upgrade farming systems, enhancing household resilience, and investing in community development.

5.2 Environmental restoration

Carbon projects in Northeast India contribute significantly to ecosystem restoration and the enhancement of ecosystem services through carbon sequestration [176]. Reforestation of jhum fallows and the creation of agroforestry plantations substantially increase carbon stocks. While conservative estimates suggest soil carbon sequestration rates of approximately 0.9–1.0 Mg C ha⁻¹ yr⁻¹ in Himalayan agroecosystems, empirical evidence from high-rainfall tropical systems indicates that sequestration rates in the North Eastern Region may range from 1.5 to 3.5 Mg C ha⁻¹ yr⁻¹ under improved land management practices [170]. In Manipur, secondary forests provide multiple co-benefits, including soil and watershed protection, flood mitigation, and the strengthening of carbon sink functions, thereby reducing soil erosion and lowering drought risks [177]. Trees in agroecosystems enhance soil structure and moisture retention, which are critical ecosystem services that improve farm productivity and strengthen climate resilience. Carbon-credit activities contribute directly to climate mitigation by expanding tree cover and increasing soil organic matter [178]. Tree-based systems in Northeast India function as significant carbon sinks and are sustainably managed to safeguard ecological integrity, aligning

with India's Nationally Determined Contributions (NDCs) and SDG 13 (Climate Action) [179]. In practice, incentivizing farmers to shift from slash-and-burn jhum cultivation to improved fallow management and mixed-tree cropping accelerates vegetation recovery and carbon uptake. In Nagaland and Manipur, 15–20 years of fallow regeneration can restore substantial aboveground biomass, in some cases reaching nearly half the carbon stock of mature forests [180]. Collectively, these interventions transform degraded landscapes into carbon-rich agroforests, delivering co-benefits such as biodiversity conservation, water security and enhanced soil fertility, while measurably reducing net CO₂ emissions.

6 Economic valuation of carbon credits

Estimating the monetary value of carbon credits requires multiplying the quantity of CO₂ sequestered by the prevailing credit price, followed by a comparison of benefits and costs using indicators such as net present value (NPV) and the benefit–cost ratio [181]. In India's evolving Voluntary Carbon Market (VCM), prices have ranged from US\$10 to 20 per tonne of CO₂ [32]. There may be potential for the Northeastern farm to sequester carbon via agroforestry and conservation practices, which can generate carbon revenue. Over a multi-year rotation, approximately 8–10 years, discounted cash flows can yield positive net present value if project costs (planting, maintenance, monitoring) are modest [182]. Cost–benefit analyses in similar agroforestry contexts report positive benefit–cost ratios greater, indicating that revenues from carbon credits, combined with co-product sales (timber, crops), exceed implementation costs [183]. It is important to note that gross revenue estimates from carbon credit generation may not fully translate into net income for smallholders. Transaction costs associated with project development, certification, third-party auditing, monitoring, and aggregator fees can reduce realized returns, particularly for small and fragmented landholdings. The following bar graph (Fig. 6) represents the potential gross income from carbon credits across selected North Eastern Region (NER) states of India. These findings suggest that carbon finance offers a substantial supplementary income opportunity for farmers and provides an economic

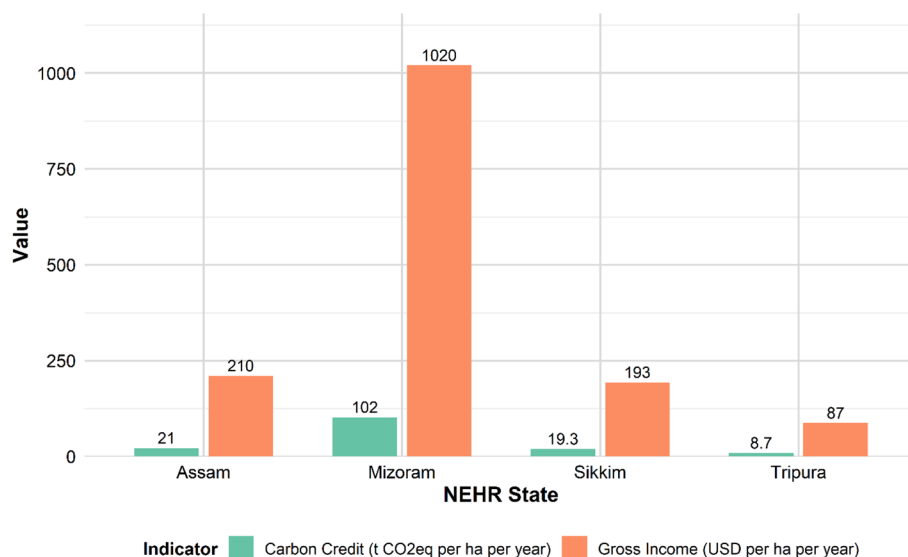


Fig. 6 Carbon credit potential and estimated gross income across states of the north eastern hill region (NEHR), India [186–189]

rationale for investment in carbon credit projects, particularly when supported by aggregation and low-cost MRV mechanisms. Carbon credit revenue estimates represent gross potential income before accounting for transaction costs, including baseline development, monitoring, reporting, verification (MRV), certification, registry fees, and administration. Empirical evidence from land-based carbon projects shows that transaction costs can substantially reduce net returns to smallholders, particularly in small-scale or fragmented systems [184, 185]. However, cooperative aggregation models and digital MRV systems can improve cost efficiency by spreading fixed verification costs across larger farmer groups.

6.1 Payment models

Under results-based carbon credit mechanisms, financial incentives to farmers and communities are conditional on the adoption of agreed-upon sustainable land-use practices, and carbon credits are formally issued only after the reported outcomes undergo rigorous, third-party verification [190]. Most projects operate through aggregation, whereby NGOs or project developers consolidate credits from thousands of smallholder farmers to achieve viable scale. Farmers may initially receive support in the form of small advances and technical inputs, such as saplings, training, and then periodic payments once credits are generated [191]. Payments are either in cash, directly to the bank accounts of farmers and community funds, or through the provision of in-kind support like farming inputs, infrastructure of an equivalent credit value [192]. For instance, a bamboo carbon project can engage villagers to plant bamboo; based on the measured growth, villagers receive yearly payments [193]. In corporate-led voluntary markets, buyers engage with carbon developers, who then pay farmers [194]. Alternative models include cooperative approaches: farmers form cooperatives or self-help groups to negotiate credit sales collectively, thereby strengthening bargaining power and reducing middlemen [195]. Government schemes, including India's sugarcane biomass programs, have experimented with premium pricing for green practices [196, 197]. Although limited formal research exists on models used in NER, effective approaches emphasize transparent accounting and timely distribution of proceeds to gain community support and ensure a fair distribution of benefits.

6.2 Internal versus external pricing mechanisms

According to Alshatri et al. [198], the market price of carbon credits can be determined by internal and external pricing mechanisms. External carbon pricing refers to selling credits on international or voluntary markets. In these markets, prices are set by global supply and demand, currently roughly \$10–40 per tonne for high-quality land-use credits [199]. Projects in NER accessing voluntary registries (Gold Standard, Verra) are influenced by external voluntary market prices, which can fluctuate but often exceed prevailing domestic carbon credit prices. Internal pricing involves India's domestic market mechanisms [200]. In 2023–2024, the Indian government established a Carbon Credit Trading Scheme (CCTS) for compliance sectors, effectively setting a base price for carbon reductions initially around Rs. 2,000 per t CO₂e [200]. Under the CCTS, voluntary projects operating on a 'baseline-and-credit' methodology, including agricultural interventions, will be formally certified, thus qualifying for the generation and trading of domestic carbon credits [201]. Thus, internally generated credits may trade at this

government-stipulated price or higher if demand exists [202]. In practice, local project developers must navigate both regimes: a developer might choose to certify on an external registry or on the new domestic platform, ensuring a fixed INR price and compliance-ready credits [148]. The key difference is that external pricing is market-driven and typically yields higher revenues in USD, while internal pricing is regulated, with prices in INR and intended to stabilize finance for local climate actions.

7 Regional synthesis: North Eastern Region (NER) of India

7.1 Potential in the NER Region of India

India's NER comprises denser forests and traditional agriculture based on tree crops [203]. Indicatively, Mizoram has around 84.5% forest cover, and Arunachal Pradesh has around 79.3 percentage [204]. Such dense forests correspond to massive biomass and soil carbon stocks. It has been estimated that, on average, long-term agroforestry can increase soil carbon by 40–43 MgCha⁻¹ through the transformation of conventional farming practices [205]. Soil organic carbon is higher in the agroforestry areas of the eastern Himalayas than in the croplands [206]. Rapidly growing trees and bamboos contribute further: *Bambusa tulda* plantations in Mizoram were found to have aboveground biomass of 36–64 MgCha⁻¹ [187]. Figure 7 exemplifies the carbon credit potential in each NER state of India. This large carbon potential indicates that the NER region has immense carbon sequestration opportunities. Increasing the area under trees, hedge-rows, and enhanced agroforestry could convert the region into a significant provider of carbon credits, owing to its naturally high biomass and soil carbon.

7.2 Emission baseline and land use patterns

The North-Eastern Hill (NER) region of India is highly forested: around 17.2 million ha, which is approximately 25% of India's forest cover, falls within this biodiversity hotspot [212]. Land cover is highly heterogeneous, including dense and secondary tropical forests [213], traditional shifting cultivation [214], fallow lands [215], tea [216], arecanut

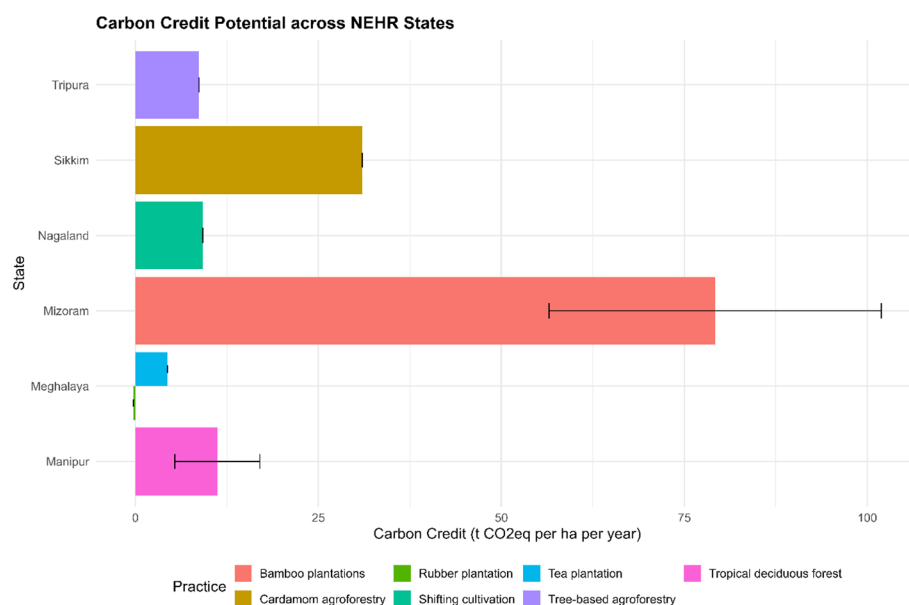


Fig. 7 Carbon credit potential across NER states [184, 185, 208–211]

[217], oil palm [218], cardamom plantations [219], bamboo stands [220], and grasslands [221]. An example of this land-use pattern is shifting cultivation fallows, which are generally slash-and-burn hill clearings that gradually build up carbon in the soil. As mentioned earlier, traditional agroforestry fields have significantly greater soil carbon than conventional farmland. Yadav et al. (2021) [222] found that a 28-year-old sissoo-pineapple agroforestry system sequestered around 71.6 MgCha^{-1} in the top 1 m of soil, compared to roughly 52.8 MgCha^{-1} in the neighbouring cropland. Nath et al. [212] demonstrated that when jhum fallows were allowed to mature for 15 to 20 years, above-ground biomass increased to around 93 MgCha^{-1} compared to 5 to 8-year-old fallows. However, even 20-year fallows were characterized by a 20–40% increase in biomass compared to nearby natural forests, indicating that traditional (smaller) fallows have a carbon deficit relative to primary forest [212]. Thus, a realistic baseline scenario for carbon credit accounting may assume that actual interventions yield lower carbon stocks than old-growth forests. Although these NE grasslands and agro-systems are degraded, they have remained net carbon sinks. Pathak et al. [223] measured net CO_2 sinks in the traditionally managed Imperata grasslands. The high forest carbon stocks offset the carbon losses that occur from land clearing and crop harvests, which define the emission baseline in the NER region. Any carbon-credit project should adjust baseline emissions to account for factors such as average jhum harvest levels and plantation clearances.

7.3 Agroforestry, Jhum, wetland rice cultivation

Although the NER region includes numerous carbon sinks, it also contains important sources of GHG emissions. For example, shifting cultivation (jhum) can be a rapid source of carbon depletion when short fallows are maintained. A simulation of Nagaland jhum by Borah et al. [208] inferred that 30-year fallows compensate for around 50% of old-growth forest carbon, whereas a 5-year cycle can result in approximately 70–80% carbon loss compared to old forests. Devi and Singh [187] found that in Mizoram, bamboo forests exhibited greater biomass and higher carbon stocks, with rapid annual carbon sequestration of $27.8 \text{ MgCha}^{-1} \text{ yr}^{-1}$ in a three-year-old *Bambusa tulda* plantation. Through wet rice cultivation in terraced systems or in Jhum cultivation, North Eastern agriculture also produces significant greenhouse gas (GHG) emissions [224]. Through livestock such as pigs, cattle, and buffalo, enteric CH_4 is emitted [225]. Livestock in the NE region is estimated to produce around $0.506 \text{ TgCH}_4 \text{ yr}^{-1}$, equivalent to 10.9 TgCO_2 [226]. Although the NER region contains substantial carbon sinks, such as agroforestry systems, and experiences significant GHG emissions from farming practices, localized data on these emissions remain limited.

7.4 Sustainable agricultural practices for carbon sequestration

In India, farmers have the opportunity to receive carbon credits by adopting practices that sequester CO_2 into long-lived carbon pools. An analysis by Cariappa et al. [227] assessed this potential within the context of voluntary carbon frameworks. Conservation agriculture in the eastern states had the potential to increase the carbon earnings of farmers by approximately $\text{US\$1,830 ha}^{-1}$. Key eligible practices include no-till (zero-tillage), retention of crop residues, and optimized fertilization. The study noted that zero-tillage wheat cultivation combined with reduced nitrogen applications had already achieved the additionality required for carbon projects.

7.4.1 Soil carbon management

The concept of soil carbon sequestration is based on enhancing soil organic carbon (SOC) through greater organic inputs and reduced soil disturbance. A 15-year mulberry (*Morus alba*) agroforestry experiment in the Himalayas demonstrated a maximum SOC accumulation of approximately $0.99 \text{ Mg Cha}^{-1} \text{ yr}^{-1}$ in the topsoil. Alternative farming practices such as cover cropping, green manuring, and conservation tillage also contribute to SOC buildup [170]. Meta-analyses indicate that cover crops, particularly when combined with no-till practices, can increase soil carbon by approximately 0.5 to $1.3 \text{ tCha}^{-1} \text{ yr}^{-1}$. Average gains of $0.4\text{--}1.0 \text{ Mg Cha}^{-1} \text{ yr}^{-1}$ SOC have the potential to accumulate substantial soil carbon stocks over decades [228]. By adopting judicious land use practices, such as planting diverse cover crops, retaining crop residues, practicing no/low tillage, and applying compost or biochar, farmers can steadily transfer atmospheric CO_2 into stable soil pools [229]. These gains are potentially verifiable for carbon credits, as measurement protocols for soil carbon change already exist [141].

7.4.2 Livestock and manure management

Livestock are a significant source of agricultural GHGs, primarily through methane produced in enteric fermentation and manure [230]. Providing high-quality feed supplements such as seaweeds and oils improves digestion, thereby reducing methane emissions from ruminant animals like cows and sheep [231]. Similarly, feeding cattle with grains rather than forage alters the fermentation process, leading to lower CH_4 emissions [232]. To mitigate enteric CH_4 emissions, feed additives (e.g., fats, oils, seaweed) have the potential to decrease emissions by approximately 30 to 40% [233]. These interventions in the NE pig, cattle, and buffalo systems could significantly reduce emissions while also contributing to clean energy transitions, like Alternate Wetting and Drying (AWD) in rice cultivation, which achieves around a 64% reduction in CH_4 [234]. Biogas plants improve manure management by preventing the direct release of methane into the atmosphere through anaerobic decomposition. This provides economic benefits through carbon credits and environmental benefits by reducing methane emissions [235]. In India, commonly used manure-based biodigesters capture CH_4 via anaerobic breakdown, simultaneously producing renewable energy and generating carbon credits [236].

7.4.3 System of rice intensification and AWD techniques

The System of Rice Intensification (SRI) and other climate-smart practices adopted in the rice cultivating areas of the North Eastern Hills (NER) hold strong potential for reducing greenhouse gas emissions [237, 238]. In SRI, young seedlings are planted at a wider spacing in moist soil [239]. This change in water and crop management substantially reduces methane (CH_4) emissions. SRI fields produce $2\text{--}13 \text{ kg CH}_4\text{ha}^{-1}\text{season}^{-1}$, which is about 60% lower than the emissions from traditional flooded systems of $37\text{--}39 \text{ kg CH}_4\text{ha}^{-1}\text{season}^{-1}$ [240]. Besides reducing emissions, SRI also enhances water-use efficiency and, in many cases, increases yields [239]. Key practices such as wider spacing, reduced water application, and earlier seedling transplantation not only lower methane emissions but also promote carbon sequestration in soils and plant biomass, thereby providing significant mitigation benefits per hectare to farmers in the NER region [242].

AWD employs minimal water and wider spacing, thereby reducing methane emissions from rice fields by $64.5 \pm 2.5\%$ and serving as both a water conservation and climate change mitigation strategy [234]. In practice, AWD involves draining the fields two to three times during the growing season, which can reduce CH_4 emissions by 60 to 65%. Both SRI and AWD not only lower emissions but also enhance productivity, thereby increasing the potential for carbon credits through the aggregation of emission reductions [234].

Enhanced rice cultivation methods, such as the System of Rice Intensification (SRI), yield the highest carbon sequestration and carbon credit values compared to conventional and zero-tillage systems [238]. SRI not only reduces greenhouse gas emissions but also increases farmers' profitability and well-being. For instance, it generates substantial revenue gains and achieves a favourable abatement cost of -2.7 VND per ton of $\text{CO}_2\text{eq ha}^{-1}$ compared to conventional farming, thereby aligning with sustainable agriculture practices and climate change mitigation efforts [243].

7.4.4 Fertilizer and nitrogen efficiency

Nitrogen fertilizers contribute significantly to nitrous oxide (N_2O) emissions, which makes it of paramount importance to enhance N-use efficiency. In resource-constrained hill agro-ecosystems such as those in the NER, low-cost nitrogen management interventions offer the most scalable mitigation pathway. Evidence indicates that site-specific nutrient management tools such as Leaf Colour Charts (LCC) significantly improve nitrogen use efficiency while reducing N_2O emissions in rice-based systems [244, 245]. Similarly, slow-release and inhibitor-treated fertilizers can reduce N_2O emissions under field conditions [246]. Precision nitrogen management in spring maize improves grain yield and nitrogen use efficiency and reduces nitrous oxide emissions by 8.0% [247], alongside intercropping with legume crops, which reduces dependency on chemical fertilizers as it naturally fixes nitrogen, thereby reducing the release of methane [35]. Using slow-release fertilizers, such as neem-coated urea and nitrogen inhibitors, will aid in the gradual conversion of ammonium to nitrate, thereby reducing its loss. Slow-release biochar-based fertilizer significantly decreases methane emissions by 33.4% in rice fields compared to mixed chemical and organic fertilizer treatments [248]. As such, by coating urea with inhibitors, urea or a polymer, nitrogen transformation can be slowed down. In a cotton field study, the addition of DCD, dicyandiamide, a nitrogen inhibitor to urea, reduced approximately 75% of N_2O emissions compared to plain urea [249]. Another field experiment carried out to assess the impact of various fertilizer types on methane (CH_4) emissions and methanogenic communities in paddy soils by Roy et al. [250] revealed that the application of biochar-based slow-release fertilizer decreased CH_4 emissions by 33.4% during the rainy season of rice cultivation, compared to mixed chemical and organic fertilizers.

7.5 Carbon sequestration potential in NER

Integrating the above practices enables NER agriculture to realize its substantial carbon sink potential. For example, a landscape-level analysis conducted in NE India calculated that the conversion of a cropland to tree-based systems adds approximately $40\text{--}43 \text{ MgCha}^{-1}$ to the ecosystem [206]. At the scale of a million hectares, this suggests that NER could sequester millions of tons of carbon dioxide through the expansion of

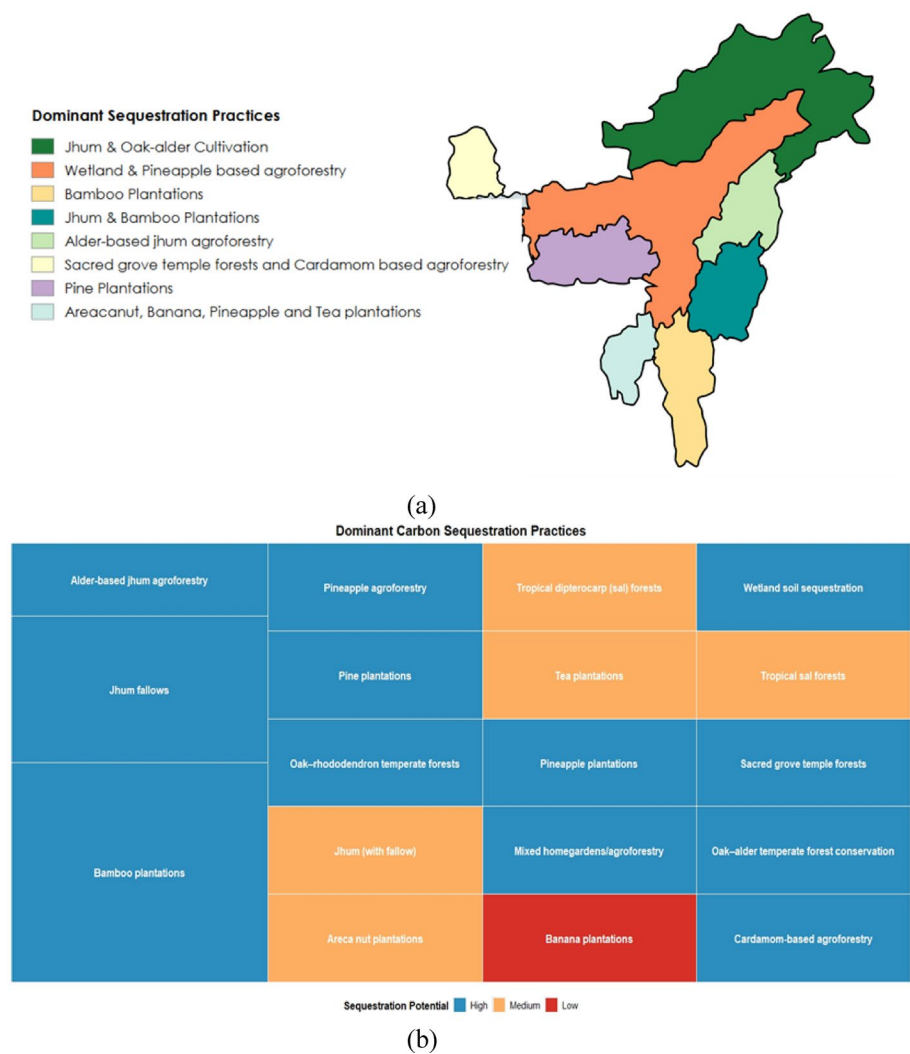


Fig. 8 a [187, 206, 207, 252, 253]. b [186–188, 210, 254]

agroforestry alone. Besides, the high soil carbon in the region (around 42–72 MgCha⁻¹ in NE forests) can lead to substantial carbon stocks being sequestered over time through modest management changes [251]. The following figure (Fig. 8a, b) illustrates the dominant carbon sequestration practices in the North East Region (NER) of India, including agroforestry, plantations, and crop management. Inclusively, there is considerable potential to expand carbon-friendly agriculture in NER, which offers significantly higher levels of sequestration compared to the national average and can result in substantial carbon credit generation.

7.5.1 Mission organic value chain development

The NER actively promotes organic agriculture and contributes to increased carbon in the soil [255]. States such as Sikkim, which has been 100 percentage organic since 2016, and Mizoram prioritize natural farming inputs such as compost, manure, and green manure over synthetic fertilizers [256]. Those organic inputs increase SOC and reduce the production of synthetic N₂O [257]. Within the framework of the Indian project “Mission Organic Value Chain Development in the North East Region” (MOVCDNER),

the transition to certified organic or natural farming is supported financially and incentivized [258]. Organic farming enhances soil carbon storage through inputs such as compost and manure [259]. These organic systems sequester more carbon than the traditional chemical-intensive farms by incorporating compost and crop rotations. Various studies in hill agriculture have found that organic and natural farming practices improve soil organic carbon (SOC) sequestration [260].

7.5.2 Local agroforestry and hedgerow integration

Many agroforestry systems in Northeastern India are already practiced by local farmers, making them feasible for carbon sequestration [261–264]. Tea plantations in the Barak Valley store large amounts of carbon in their biomass, with an average biomass of around 75 Mgha⁻¹, corresponding to 35–38 Mg of C ha⁻¹ [265]. Similarly, oil-palm agroforestry shows latent potential, as the combined planting of food crops (turmeric, maize, pineapple) with oil palm on degraded jhum land helps enhance carbon sequestration. The oil palm-pineapple system was estimated to have a total ecosystem carbon stock of 55.4 MgCha⁻¹, which is significantly higher than that of oil palm monoculture [266]. Parmar et al. [267] found that integrating hedgerows and *Alnus nepalensis* strips into the maize–mustard–potato cropping system increased SOC by up to 60% compared to other systems. Cover cropping with legumes also significantly enhanced soil carbon storage, with cover crops alone adding 0.5 to 1.3 tCha⁻¹ yr⁻¹ [228]. They also noted that when cover cropping practices are integrated with no-till farming, carbon sequestration could increase to 1.4 tCha⁻¹ yr⁻¹. This highlights why the combination of hedgerow systems and agroforestry holds significant strategic potential for scaling up carbon sequestration, making it a critical pathway for operationalizing the carbon credit mechanism and advancing carbon–neutral farming systems in the NER.

7.6 Case projects

7.6.1 REDD+ implementation in NE states

REDD+ refers to “Reducing Emissions from Deforestation and Forest Degradation,” with the ‘plus’ indicating additional activities such as forest conservation, sustainable forest management, and the enhancement of forest carbon stocks in developing countries [268]. REDD+ was enacted by the ministry of environment, forest and climate change (MoEFCC) in 2018, emphasizing forest monitoring, institutional frameworks for carbon accounting, and benefit sharing [269]. Its implementation is facilitated by the Forest Rights Act (FRA, 2006) and the National Working Plan Code (2014), which safeguard community forest rights and promote sustainable forest management. The main goal of the REDD+ program, aligned with the United Nations Framework Convention on Climate Change, is to support national government initiatives aimed at reducing human pressure on forests, which contributes to greenhouse gas emissions at the national level. Globally, many such initiatives are expanding, emphasizing agroforestry and addressing climate change through adaptation and mitigation strategies [270]. Voluntarily traded carbon credits are a major source of financing for Reduced Emissions from Deforestation and Forest Degradation (REDD+) initiatives in developing countries [271]. They also support carbon-sequestration-oriented projects across sectors such as agriculture, renewable energy, and waste management. Forest-rich states in the North Eastern Region (NER) have significant potential to pilot REDD+ carbon credit projects.

7.6.2 Khasi Hills and Eka project

The Eka Project, initiated by EKI Energy Services Ltd. (formerly EnKing International), is a large-scale initiative aiming to generate one billion carbon credits by 2027 and achieve net-zero emissions by 2030 [272]. The project incorporates community-based solutions, renewable energy installations, nature-based interventions, and investments in green technologies. By generating verified carbon credits, it strategically leverages carbon market mechanisms to support India's nationally determined contributions (NDCs) under the Paris Agreement. The initiative integrates social, environmental, and economic objectives, paving the way for sustainable development [273]. Brahma et al. [176] found that converting degraded forests and grasslands into managed plantations in Northeast India significantly increased ecosystem carbon stocks. In the NER region of India, traditional biomass-based cooking methods are prevalent, contributing to deforestation and indoor air pollution. A key component of the Eka project is the deployment of the improved cookstoves (ICS), which reduce firewood consumption and associated CO₂ emissions.

8 Constraints on carbon credit

Despite their potential, carbon credit mechanisms face several structural and operational challenges that hinder effective implementation. These constraints arise from both the supply and demand sides of the carbon credit market. Key issues include inconsistent measurement, reporting and verification (MRV) protocols; policy and regulatory limitations; economic and market uncertainties; limited farmer participation due to insufficient incentives; and various technical and biophysical challenges [274]. On the demand side, challenges include low buyer confidence stemming from complex, non-transparent carbon accounting procedures, which cast doubt on the credibility of issued credits [275]. Certification schemes such as VERRA, Gold Standard, Plan Vivo, and Puro Earth can create confusion among buyers interested in carbon markets [276]. Consequently, the market price of carbon credits remains relatively low, limiting their attractiveness to both producers and investors [277]. On the supply side, the low potential for carbon credit generation from small landholdings, insufficient farmer awareness and training, lack of connection with carbon companies, and non-receipt of promised carbon credit payments discourage continued participation [278]. The risk of reversibility of captured SOC persists [279]. Furthermore, atmospheric temperature, precipitation, soil properties, topography, and variations in vegetation contribute to high spatial variability in soil carbon storage [280].

8.1 MRV and smallholder participation

Northeast India faces significant hurdles in carbon market participation due to limited and imprecise measurement, reporting, and verification (MRV) tools for quantifying carbon [281], as well as a lack of local data and skilled experts [282]. Delays in verification processes further postpone farmer payments, while intermediaries often claim a disproportionate share of revenues. Field surveys are often expensive for data collection, and satellite-based monitoring is hindered by adverse weather conditions and persistent cloud cover [283]. The quantification of SOC using prediction models for a particular soil type or vegetation may be inaccurate, especially for small and marginal farmers. Smallholders in the NER region face considerable challenges in engaging with voluntary

carbon markets, including complex certification protocols, high transaction costs, and a lack of regional aggregators or project developers [284, 285]. Project validation and documentation are required for most projects and are usually carried out by remote companies, limiting local participation [286]. In practice, smallholder participation is constrained by high transaction costs, difficulties in aggregation, fragmented landholdings, complex tenure arrangements, delayed payments, and asymmetric power relations between farmers and intermediaries. These factors significantly influence participation incentives and distributional outcomes.

8.2 Institutional and tenurial complexity

A significant institutional constraint to the implementation of agricultural carbon credit projects in the North Eastern Region (NER) of India arises from the prevalence of communal and customary land tenure systems. In several NER states, including Meghalaya, Nagaland, Mizoram, and parts of Manipur, land is traditionally owned and managed by clans, tribes, or village councils rather than by individual households. Such collective tenure arrangements, while socially embedded and culturally resilient, pose legal and contractual challenges for carbon credit mechanisms that typically require clearly demarcated land parcels, identifiable project proponents, and long-term permanence commitments extending 20–30 years. The absence of formal individual land titles complicates project registration, benefit attribution, and enforcement of permanence obligations under prevailing voluntary and compliance carbon market standards. Addressing this challenge will require the development of community-level contracting frameworks, aggregation through customary institutions, and formal recognition of collective ownership structures within carbon registries. Aligning carbon market rules with customary governance systems is therefore essential to ensure both environmental integrity and social legitimacy of carbon credit initiatives in the NER. Research on community forest governance in Northeast India indicates that customary tenure systems are deeply embedded in clan-based and village institutions, where collective decision-making governs land allocation and forest management [287]. Studies further show that secure community tenure enhances long-term conservation outcomes and carbon stock stability [288–290]. Therefore, carbon credit mechanisms that require 20–30-year permanence commitments must be structured through community-level contracts and benefit-sharing agreements rather than individual land titles. Aligning permanence provisions with customary tenure structures is essential for ensuring project durability and reducing reversal risk.

9 Conclusion

This study is novel in that it integrates carbon market frameworks, biophysical carbon quantification models, and region-specific community-led case evidence to present the first consolidated analytical roadmap for operationalizing agricultural carbon credits in India's North Eastern Region. The advent of carbon credit markets represents a significant policy and economic innovation to incentivize sustainable agricultural practices by converting ecological stewardship into tradable economic assets. By monetizing carbon sequestration and emissions reductions, these markets provide a financial rationale for climate-smart land management. Biophysical models such as DNDC and CENTURY introduce scientific rigor into the measurement, reporting, and verification

(MRV) of agricultural carbon outcomes, thereby enhancing transparency and credibility. In India, emerging carbon market policies seek to balance national climate commitments under the Paris Agreement with the objective of improving farmer livelihoods. The North Eastern Region (NER) possesses distinctive potential for agricultural carbon credit generation owing to its rich biodiversity, extensive agroforestry systems, and traditionally low-input farming practices. Pilot initiatives such as REDD+ programs and the Eka Project demonstrate that indigenous and community-managed landscapes can participate meaningfully in international carbon markets while advancing the United Nations Sustainable Development Goals (SDGs). These initiatives generate multiple co-benefits, including supplementary farm income, strengthened local institutions, biodiversity conservation, and enhanced climate resilience. Carbon credits therefore, extend beyond financial returns to encompass ecosystem services such as soil health improvement, watershed regulation, and long-term climate stability. By linking farmers to carbon markets, new livelihood opportunities emerge through MRV services, agricultural extension, aggregation platforms, and climate-smart value chains. Despite this potential, structural challenges persist in the NER, including weak institutional capacity, high transaction costs, fragmented landholdings, customary land tenure systems, and limited awareness among smallholders. Without deliberate policy and institutional design, these constraints risk excluding small and marginal farmers from carbon market participation. Addressing these challenges requires a phased and region-specific implementation pathway.

- I. Short-term (0–3 years): Capacity building, carbon literacy training, and aggregation through Farmer-Producer Organisations (FPOs) and customary village institutions.
- II. Mid-term (3–7 years): Deployment of digital MRV tools integrating remote sensing and process-based models to reduce transaction costs and improve verification efficiency.
- III. Long-term (7–15 years): Gradual integration of agricultural carbon credits into India's emerging compliance carbon market framework, ensuring alignment with permanence safeguards and customary tenure governance structures.

Such a phased approach is likely to improve scalability, institutional equity, and environmental integrity within the regional carbon framework. In this context, carbon credits may be interpreted not only as tradable financial instruments but also as policy mechanisms capable of influencing broader agricultural and rural development processes. By acknowledging farmers as contributors to ecosystem service provision and incorporating indigenous land-use systems within formal climate finance structures, carbon market participation has the potential to support more resilient, regenerative, and inclusive agricultural systems in the North Eastern Region of India.

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Author contributions

Shanmugam Gokul: conceptualization, visualization, writing—original draft preparation. Thangavel Pradeesh Kumar: conceptualization, writing—reviewing and editing, funding acquisition. Ramasubramanian Sabarivasan: visualization, writing—reviewing and editing. Lavanya Lahari Soprala: writing—reviewing and editing.

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Data availability

No datasets were generated or analysed during the current study.

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Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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